Bayesian Learning Exam

The following libraries gets installed automatically when you render this quarto document:

- `mvtnorm`, for the multivariate normal density and random number generator.

> You can use **chunk options** to control if a chunk is run or not (useful if the code errors so that the document does not compile. You can still get partial points if the code is showing):
>
> - Not running the code: `#| eval: false`
> - Not showing code: `#| echo: false`
> - Not showing output: `#| output: false`
>
> Example code chunk that shows the code, but the code is not run since it has an error:
>
> ```
> my_studid_variable = 4
> my_undefined_variable_that_errors
> ```

Problem 1

```
load("reaction.RData")  # loads the vector y with 84 reaction times
```

Problem 2

This problem is solved and handed in on paper.

Problem 3

```
load("reaction.RData")  # loads the vector y with 84 reaction times
# The skew normal density function
dskewnormal <- function(y, alpha = 0, log = FALSE){
  if (log == FALSE){
    return(2*dnorm(y)*pnorm(alpha*y))}
  else{
    return(log(2) + dnorm(y, log = TRUE) + pnorm(alpha*y, log.p = TRUE))
  }
}
```

Problem 4

```
# Nothing to load for this problem.
```